Animesh Choudhury

PhD Researcher in Earth & Atmospheric Sciences | ML for Weather Forecasting | Remote Sensing Specialist

PROFESSIONAL SUMMARY

PhD researcher with 6+ years of experience in Earth and Atmospheric Sciences, specializing in data-driven weather forecasting, machine learning for climate science, and geospatial analysis. Currently developing cutting-edge weather prediction systems using Pangu-Weather architecture and IMDAA reanalysis data at NIT Rourkela. Proven track record with publications (120+ citations) in high-impact journals including Remote Sensing, Journal of Environmental Management, and Urban Climate. Expert in WRF mesoscale modeling, deep learning for atmospheric sciences, drought monitoring, and satellite remote sensing with proficiency in Python, R, Google Earth Engine, and advanced ML frameworks.

EDUCATION

PhD in Earth and Atmospheric Sciences | 2022 - Current

National Institute of Technology, Rourkela, India

Post Graduate Diploma in Remote Sensing & GIS | 2017-2018

Indian Institute of Remote Sensing (IIRS), Dehradun, India

M.Sc in Environmental Science | 2015-2017

Asutosh College, Calcutta University, Kolkata, India

B.Sc in Physics | 2009-2013

Raghunathpur College, Burdwan University, West Bengal, India

WORK EXPERIENCE

Junior Research Fellow (PhD Researcher)

National Institute of Technology, Rourkela, India | 15/02/2022 - 30/06/2025

- Developing data-driven weather forecasting systems using Pangu-Weather architecture and IMDAA reanalysis data
- Creating BharatBench dataset for ML-based weather forecasting over India
- Conducting high-resolution mesoscale modeling of urban thunderstorms using WRF
- Analyzing land-atmosphere interactions and urban heat island effects

Junior Research Fellow

Department of Remote Sensing and GIS, Vidyasagar University, Midnapore, India | 21/01/2020 - 07/02/2022

- Long-term monitoring of droughts using satellite-based indices (SPEI, VCI, TCI)
- Developed ML models for prediction of agricultural droughts
- Analyzed regional variation of drought parameters across India
- Published research in Journal of Environmental Management (203 citations)

Junior Research Fellow

DST-Mahamana Centre of Excellence in Climate Change Research, Banaras Hindu University, Varanasi, India | 13/06/2019 - 19/01/2020

- Observation of horizontal and vertical distribution of air pollutants using CALIPSO LIDAR
- Analyzed aerosol transport patterns over Indo-Gangetic plain
- Integrated ground-based and satellite observations for air quality assessment

Junior Research Fellow

Indian Institute of Remote Sensing, Dehradun, India | 21/10/2018 - 30/03/2019

- Snow cover area mapping of Northwest Himalaya using MODIS, Sentinel, Landsat datasets with Google Earth Engine
- Data collection, installation, and supervision of automatic weather station (AWS), snow precipitation gauge (SPG), and snow pack analyzer (SPA)
- Published research in Remote Sensing journal (145 citations)

Junior Research Fellow

CSIR - Institute of Himalayan Bioresource Technology, Palampur, India | 12/08/2018 - 14/10/2018

- Collection of spectral signatures using ASD Fieldspec spectroradiometer
- Correlated spectral data with bio-physical parameters
- Classification of tree species using satellite-based hyperspectral data

SELECTED PUBLICATIONS

[1] Panda, J., Mukherjee, A., Sarkar, A., Kumar, G., & Choudhury, A.. **Analysis and Prediction of PM2.5 Concentrations Over Indian Megacities Using Geospatial Techniques and Deep Learning Models**. *Geospatial Technologies*, 2026.

[2] Roy, D., Panda, J., Mukherjee, A., Choudhury, A., & Panda, S. K.. **A long-term analysis of pre-monsoon thunderstorm characteristic features over major urban areas of India**. *Acta Geophysica*, 2026.

[3] Choudhury, A., & Panda, J.. **Development of a Data-driven weather forecasting system over India with Pangu-Weather architecture and IMDAA reanalysis Data**. *In Preparation*, 2025.

[4] Choudhury, A., Panda, J., & Mukherjee, A.. **BharatBench: Dataset for data-driven weather forecasting over India**. *In Preparation*, 2024.

[5] Mukherjee, A., Panda, J., Choudhury, A., Singh, S., & Bhattacharyya, S.. **Analyzing urban footprints over four coastal cities of India and the association with rainfall and temperature using deep learning models**. *Urban Climate*, 2024.

[6] Singh, S., Mukherjee, A., Panda, J. et al.. **Analysis and Forecasting of Temporal Rainfall Variability Over Hundred Indian Cities Using Deep Learning Approaches**. *Earth Systems and Environment*, 2024.

[7] Choudhury, A.. **Drought trend and its association with land surface temperature (LST) over homogeneous drought regions of India (2001–2019)**. *Discover Water*, 2024.

[8] Panda, J., Mukherjee, A., Choudhury, A., Biswas, S.. **Urban Heat: UHI and Heat Stress Threat to Megacities**. *Climate Crisis: Adaptive Approaches and Sustainability. Sustainable Development Goals Series. Springer, Cham*, 2023.

[9] Budakoti, S., Singh, C. & Choudhury, A.. **Transport of a severe dust storm from Middle East to Indian region and its impact on surrounding environment**. *International Journal of Environmental Science and Technology*, 2022.

[10] Choudhury, A., Dutta, D., Bera, D., Kundu, A.. **Regional variation of drought parameters and long-term trends over India using standardized precipitation evapotranspiration index**. *Journal of Environmental Management*, 2021.

[11] Choudhury A, Yadav AC, Bonafoni S. **A Response of Snow Cover to the Climate in the Northwest Himalaya (NWH) Using Satellite Products**. *Remote Sensing*, 2021.

CONFERENCE PAPERS & PROCEEDINGS

[1] **A Step Towards Building Medium-Range Data-Driven Weather Forecasting System Over India**. *IEEE Conference*, 2024.

[2] **Monitoring of Surface Water of East Kolkata Wetlands (1984-2015) with Global Surface Water Dataset**. *ACRS 2020*, 2020.

[3] **Role of Earth Observation Data and Hydrological Modeling in Supporting UN SDGs in North West Himalaya**. *ISPRS Archives*, 2020.

[4] **Simulation of an Extreme Dust Episode using WRF-CHEM**. *IASTA 2018*, 2018.

TECHNICAL SKILLS

Weather & Climate Modeling: WRF-CHEM, Pangu-Weather, Mesoscale Modeling, IMDAA Reanalysis, Climatological Data Analysis

Machine Learning & AI: Deep Learning, TensorFlow, PyTorch, Time Series Forecasting, Neural Networks, Data-Driven Models

Remote Sensing: MODIS, Sentinel, Landsat, SNAP, ENVI, Idrisi, Hyperspectral Analysis, ASD Fieldspec

GIS & Geospatial: Google Earth Engine, ArcGIS, QGIS, Spatial Analysis, Geospatial Modeling

Programming & Data Science: Python, R, SQL, Data Processing, Data Analysis, Data Visualization

Climate Analysis: Drought Monitoring, SPEI/VCI/TCI Indices, Climate Trends, Snow Cover Mapping, Atmospheric Science

POSTER, ONLINE TRAINING, WEBINAR

- **Poster Presentation at AOGS 2025** on "Regional Data-driven Weather Forecasting Over India with Pangu-weather Architecture and IMDAA Reanalysis" in the 22nd Annual Meeting of Asia Oceania Geosciences Society held at Sands Expo & Convention Centre Marina Bay Sands, Singapore during 27 Jul - 01 Aug, 2025
- **National Symposium Participation** in the Tropment-2024, Indian Meteorological Society held at NIT Rourkela during 10-12 December 2024
- **Organizing Member and Speaker** in the National Workshop on Drought Monitoring and Early Warning using Geospatial Techniques held at Department of Remote Sensing and GIS, Vidyasagar University during 24th February, 2021
- **Poster Presentation at AAPW-5** on "Aerosol vertical distribution over lower Gangetic plain using CALIPSO LIDAR" in the Asian Air Pollution Workshop during 5th-7th November, 2019
- Completed the "Recent Trends in Time Series Forecasting using Deep Learning Models" organised by Department of Computer Science & Engineering, NIT Rourkela during December 25-29, 2023
- Completed the "Deep Learning Application for Smart Cities" organised by Department of Computer Science & Engineering, NIT Rourkela during September 11-15, 2023
- Completed the "Machine Learning in Weather & Climate (MOOC)" organised by ECMWF in partnership with IFAB during January to April, 2023
- Completed the "Capacity Building - Training School on WRF Modeling System" organised by C-DAC and SAMA under National Supercomputing Mission (NSM) during 3rd to 20th April 2023
- Completed the "Open Classroom Online Lecture Series in Atmospheric and Climate Sciences" organised by SAMA and Centre for Atmospheric Sciences, SRM Institute during 7-24 April 2023
- Completed the "Geospatial Modeling Driven Urban Hazard and Risk Analysis" organised by IIRS, Dehradun during 13-02-2023 to 17-02-2023
- Completed the "Machine Learning (ML) and Deep Learning (DL) for Earth Observation (EO) Data and Application" organised by DA-IITC in Collaboration with SAC, ISRO during 9th-13th August, 2021
- Attended a webinar on "Environment and Ecosystem Restoration" organized by Department of Physics, K.L.E.F., Vijaywada, Andhra Pradesh on 5th June, 2021
- Attended a webinar on "Geospatial Techniques in drought characterization and impact assessment" organized by Albedo Foundation on 26th September, 2020

Last updated: March 2026